

1. What is a token, and how are tokens used to control network access?

A special control message is called a *token*.

In a token passing system, when no station has any packets to send, the token circulates among all stations continuously.

When a station captures the token, it sends its data, and when transmission completed, it releases the token.

2. Distinguish between adaptive and non adaptive routing algorithms. Why is adaptive routing superior to non adaptive routing?

Non adaptive Routing:

Once a pathway to a destination has been selected the router sends all packets for that destination along that one route.

The routing decisions are not based on the conditions or topology of the networks.

Adaptive Routing:

Router may select a new route for each packet. (even packets belonging to the same transmission)

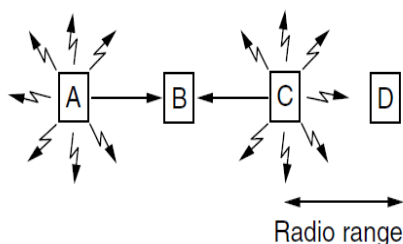
The routing decisions are based on the conditions or topology of the networks.

Adaptive routing is superior to non adoptive routing because adaptive routing may select a new route for each packet in response to change in condition and topology of the networks.

3. Briefly describe the hidden and exposed station problems in wireless LAN.

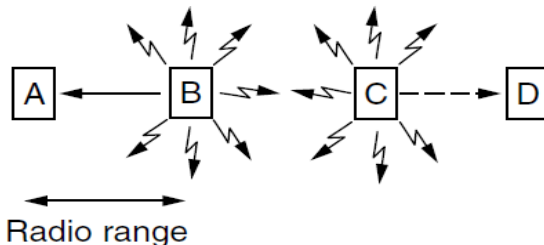
Hidden terminals where two stations may be able to both communicate with a particular destination, but those stations may not be able to communicate with each other. So these terminals are senders that cannot sense each other but nonetheless collide at intended receiver.

- A and C are hidden terminals when sending to B



Exposed terminals are senders who can sense each other but still transmit safely (to different receivers)

- Desirably concurrency; improves performance
- $B \rightarrow A$ and $C \rightarrow D$ are exposed terminals



The resolution to these problems is to use CSMA/CA (Carrier Sense Multiple Access / Collision Avoidance) as opposed to CSMA/CD in this case.

4. Briefly describe the term “*cache coherency*”.

Data in a computer with cache memory may be in the cache and memory.

Clearly, it is important that the data in both memories should be the same or, if the values differ, the older and incorrect or stale value should be marked as invalid. In a single processor system, this is not normally a problem. However, with multiple processors, each with its own private cache, it is possible for one processor to update its cache and the (same) data in the other processor’s cache will remain the same (and be stale).

Consequently, computers with multiple processors and caches must ensure that when a processor updates its own cache, all other lines containing the same data must either be updated or invalidated. The process of ensuring that the data remains in step is called cache coherency.

5. (i) What is a data hazard? Name and describe a hardware technique that overcomes most data hazards.

A data hazard is when the ALU requires a value that is not in the register file where it is supposed to be.

The hardware technique known as EX Forwarding can overcome this problem by moving the result from the first instruction to the second straight from the EX cycle, bypassing the need to wait for the MEM and WB stages.

This eliminates the need for stalls in the pipeline.

(ii) Consider the following generic codes.

```
LDR r1,[r2]
LDR r3,[r4]
ADD r6,r1,r3
SUB r7,r1,r3
MUL r3,r6,r7
STR [r2],r3
```

Identify all the potential data hazards in the above codes.

```
LDR r1,[r2]
LDR r3,[r4]
ADD r6,r1,r3 RAW on r1 and r3
SUB r7,r0,r3 RAW on r3
MUL r3,r6,r7 WAR on r3
STR [r2],r3 RAW on r3
```

6. A processor executes an instruction in the following six stages. The time required by each stage in picoseconds (1,000 ps = 1 ns) is given for each stage.

IF	instruction fetch	300 ps
ID	Instruction decode	150 ps
OF	Operand fetch	250 ps
OE	Execute	350 ps
M	Memory access	700 ps
OS	Operand store (write back)	200 ps

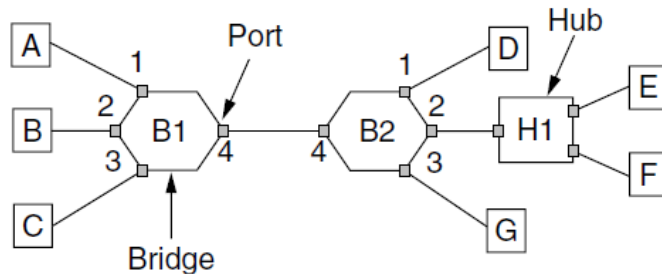
- (i) What is the time to execute an instruction if the processor is not pipelined?

$$\begin{aligned} \text{Add up the individual times: } & 300 + 150 + 250 + 350 + 700 + 200 \\ & = 1950\text{ps} = 1.950\text{ns} \end{aligned}$$

- (ii) What is the time taken to fully execute an instruction, assuming that this structure is pipelined in six stages and that there is an additional 20 ps per stage due to the pipeline latches?

The longest stage is 700 ps which determines the clock period.
With 20 ps for the latches, the time is $720 \times 6 = 4320 \text{ ps} = 4.32 \text{ ns}$.

7. Consider the following bridged LAN connected using bridges *B1* and *B2*. Suppose the hash tables in the two bridges are empty. List all ports on which a packet will be Forwarded for the following sequence of data transmissions:



- (i) A sends a packet to C.
- (ii) E sends a packet to F.
- (iii) F sends a packet to E.
- (iv) G sends a packet to E.
- (v) D sends a packet to A.
- (vi) B sends a packet to F.

- (i) B1 will forward this packet on ports 2, 3, and 4. B2 will forward it on 1, 2 and 3.
- (ii) B2 will forward this packet on ports 1, 3, and 4. B1 will forward it on 1, 2 and 3.
- (iii) B2 will not forward this packet on any of its ports, and B1 will not see it.
- (iv) B2 will forward this packet on port 2. B1 will not see it.
- (v) B2 will forward this packet on port 4 and B1 will forward it on port 1.
- (vi) B1 will forward this packet on ports 1, 3 and 4. B2 will forward it on port 2.

8. Briefly explain the operations of a *transparent bridge* and list any two (2) functions which a bridge can not perform.

Transparent bridges require no initial programming. After installation on a network, they "learn" the location of network devices by associating the source address in incoming packets with the line they were received on. Using these relationships, transparent bridges build tables of device/segment pairs. Bridges can then forward packets appropriately by analyzing a data packet's destination address. If the destination address is on the segment from which the packet came, the bridge discards it. If not, the packet is sent out on the appropriate line.

Bridge cannot determine most efficient path.

Traffic management function.

10. Briefly describe the operations of *CSMA/CD* protocol used in Ethernet.

Carrier sense multiple access with collision detection (CSMA/CD) is a form of medium access control in which a station listens to the medium to try to see if another transmission is in progress.

If the medium appears idle, the station transmits. If a collision occurs, the workstations wait a random amount of time before trying again.

CSMA/CD types: Non-persistent, 1-persistent and P-persistent:

11. Why does *CSMA/CD* use a random delay?

Randomization is used to avoid having multiple stations transmit simultaneously as soon as the cable is idle. That is, the standard specifies a maximum delay, d , and requires each station to choose a random delay less than d after a collision occurs.

In most cases, when two stations each choose a random value, the station that chooses the smallest delay will proceed to send a packet and the network will return to normal operation.

12. (a) Briefly describe the term "Instruction pipelining".

The fetch, decode and execute MEM & WB cycles for several instructions are performed simultaneously to reduce overall processing time. This process is referred as instruction pipelining.

- (b) What do you understand by the term "data dependency"?

Data dependencies occur in an executing program when, because of latency of a pipeline, the program may not execute correctly.

- (c) Briefly describe the three classes of data dependencies that could occur during program execution.

Read after write (RAW), also known as a true dependency.

Write after Read (WAR), also known as antidependency.

Write after Write (WAW), also known as output dependency.